



Module 13

Training Program Structure

At the core of any effective training program is structure. A popular and well-known training structure is **periodization**. Periodization involves a progression of several cycles of training periods to get an individual to peak performance level by a particular date or time. Each training period has a specific focus (e.g., increasing aerobic capacity). The key element of periodization is that each training period builds on the prior training period. Periodization also focuses on alternating phases of stress and recovery to allow for the desired physiological training adaptations to occur. While periodization is most often applied to training programs that are at least one year in duration, it is also applied to event-specific training programs. Periodization has garnered its fair share of criticism as being rigid, outdated, and flawed.

The polar opposite of periodization is “winging it” and randomly assigning workouts without much, if any, reasoning behind them. While this sounds crazy, this is often the training structure chosen by coaches who lack the knowledge of how the body functions. More often than not, these coaches apply the same training standards, structure, and protocol to all of their clients. The advice and programs given to these clients are often based off what the coach personally does or what he or she recently read in a magazine. This, of course, is not the advised way to structure a training program.

What Is the Right Approach?

This is the million-dollar question that has not and likely cannot be answered because of the fact that the “correct” program structure is based on the individual needs of a client. That said, as with most things related to human performance, the optimal training structure can likely be represented as a bell curve – as noted in figure 13.1.

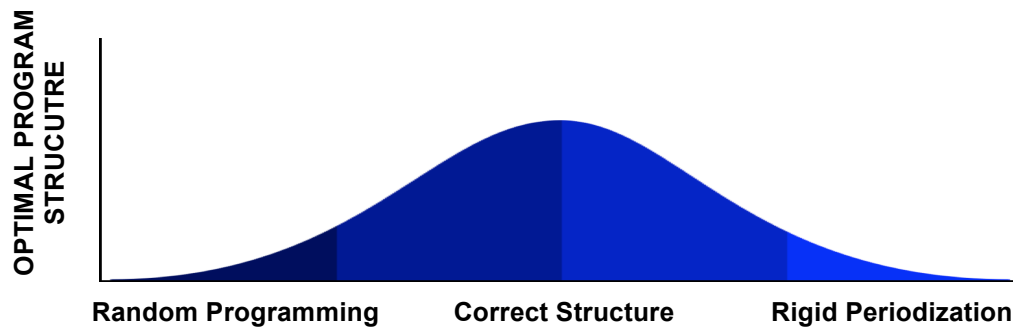


Figure 13.1 Optimal training structure bell curve

As you can see from figure 13.1, the correct structure is not a finite point on the bell curve but rather a general area – meaning there is quite a bit of room for interpretation and customization.

The correct training structure must be based on an individual, not a predefined set of parameters. While it is fine to have phases within a training program that place an emphasis on a particular type of physiological adaptation, these phases should not be rigid and should exist more to provide general direction to a training plan than strictly dictate its content.

Periodization is a polarizing topic. As such, this module will discuss the different models of periodization as well as the pros and cons.

❖ OBJECTIVES

Upon completion of this module, you should have an understanding of the following areas:

- General adaptation syndrome (GAS)
- Periodization models
 - Classic
 - Block
 - Hybrid
- Progression of periodization models
- Dealing with missed training days
- Issues with program adherence

- Training themes
- Tapering
- Peaking
- Overall trends in periodization program design
- How to find balance when creating a training program
- Reasons for criticism and proposed solution

❖ GENERAL ADAPTATION SYNDROME

To understand how the body becomes stronger, it is important to understand how the body adapts to stress. This can be traced back to findings from Hans Selye, an endocrinologist who found that when stressors were placed on mice in a laboratory setting, all the mice demonstrated similar response levels. This adaptation is referred to as the **general adaptation syndrome (GAS)** (96).

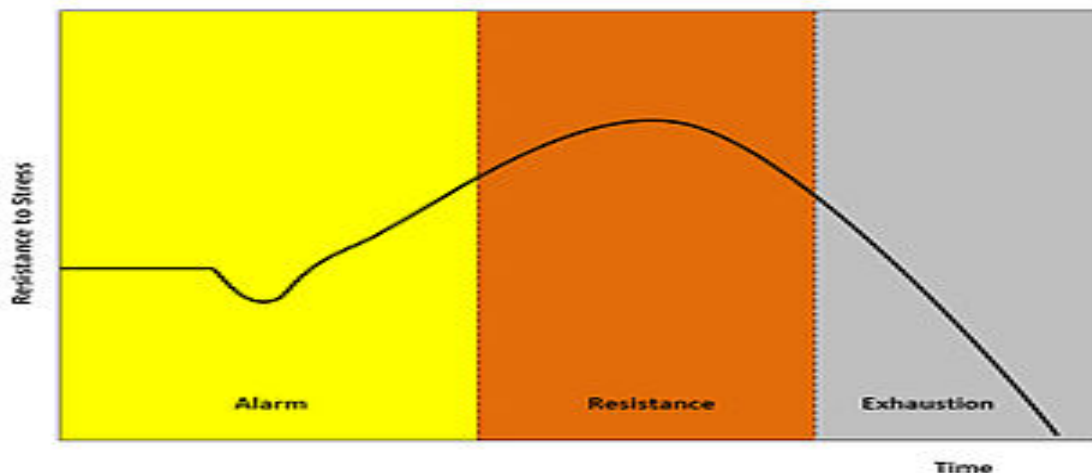


Figure 13.2 General Adaptation Syndrome (519)

Myers, D. (July 08): Exploring Psychology, 7th Edition (Worth). Pg. 398. Creative Commons Attribution 3.0

GAS comprises three stages (96):

1. **Alarm:** When alarmed, the body has an immediate response of fight or flight where it shifts its resources to muscular and emotional needs.
2. **Resistance (also called *adaptation*):** The body is able to adapt to stressful situations and become stronger.

3. Exhaustion: If high levels of stress persist for too long, the body will begin to break down.

When a runner trains or races, the body recognizes the effort as a stressor and it adapts to deal with the stress. The goal of an athlete is not to end up in the third category of GAS: exhaustion (i.e., overtraining). Overtraining occurs when an individual continues to put stress on the body without adequate rest. As a result of this, the body will eventually break down and performance will decrease. This is a crucial aspect of effective endurance-sports training, as athletes respond differently to the stressors put on them. For example, some athletes respond best to multiple back-to-back hard training days while other athletes respond best to at least one recovery day in between hard training days.

It is important to note that the resistance stage of GAS differs in time and intensity depending on the athlete. For example, a professional runner would likely have a longer duration in the resistance phase than a recreational runner, and the professional would certainly have a higher effort level.

The concept of having at least one day off between exercising the same muscle group via strength training is standard. What is not as well-known is how many days an endurance athlete should do cardiovascular training before taking a day off or adding an easy day. To answer this question, two primary factors need to be considered.

1. Intensity and/or distance of the training session
2. Recovery rate of the athlete

Generally speaking, the harder the intensity or the longer the distance of a training session, the more rest that is required. The recovery rate of an athlete is largely influenced by level of fitness. The better conditioned an athlete is, the faster the recovery rate will typically be.

The optimal amount of rest required between training sessions is best ascertained through trial and error. The most important thing to be aware of is that athletes will respond differently to stress, and therefore training programs must reflect this.

Below is a diagram (figure 13.3) that illustrates the basic concept of stress and recovery in relation to the construction of a training program.

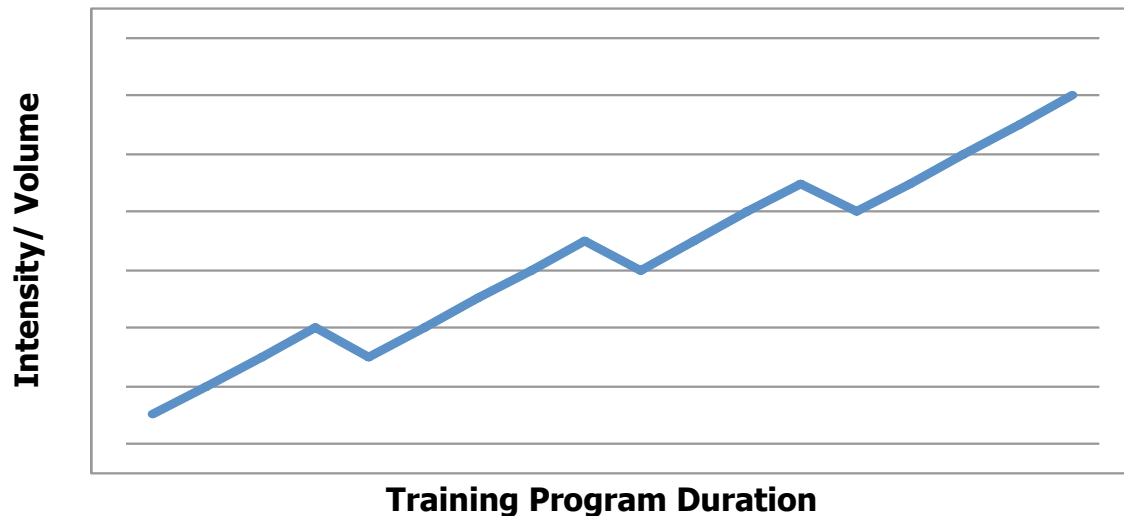


Figure 13.3 Stress/recovery structure

Rest periods (decrease in volume/intensity) are integrated into a program to allow the body to recover and profit from periods of stress (i.e., intensity/volume). This, in turn, enables a client to attain a higher level of fitness. This results in further increases in volume and intensity.

EASY BEFORE HARD

While training programs differ in their construction, a general theme is that training programs should progress from easy to hard, both in regard to intensity and distance. As noted previously, a quote by Jay Johnson sums up the rationale behind this training practice:

“Metabolic fitness precedes structural readiness.”

This means that individuals’ cardiovascular adaptation occurs at a faster rate than their musculoskeletal adaptation. As an example, from a cardiovascular standpoint, a runner might have no problem running five miles at a six-minute-mile pace. However, the person’s muscles, bones, and connective tissue might not be able to sustain this effort. It is theorized that this is a reason for many sports injuries, especially running.

The overall trend when creating a training program should be to focus on proper progression. This is not to say that hard efforts cannot or should not be fused into the initial stages of a training program. However, they should be integrated intelligently.

MOUNT EVEREST ANALOGY

Perhaps the best analogy regarding the theory behind the creation of a training program is the method used to climb Mount Everest. Because of the elevation, the mountain is climbed in stages. This allows climbers to acclimate properly to the altitude. Like altitude acclimatization, training program progression is done in stages to properly progress athletes to their goal. If athletes were to progress too fast, they would risk injury and lack the required fitness level to perform on race day.

In mountaineering, there is an elevation termed the “death zone” (higher than 26K feet). At this point, the human body can no longer acclimate to the extremely low level of oxygen (313). Humans can stay at this elevation only for short periods before being forced to descend to a lower altitude ... or die. The death zone is analogous to maintaining too high a volume/intensity based on the fitness level of a client.

While death is not the result (let's hope not!), trying to expand the time spent doing high mileage/and, or intensity beyond what the body can handle at a given point may result in long-lasting effects, including injury and extreme fatigue (i.e., overtraining).

❖ PERIODIZATION MODELS

The concept of periodization was developed during the 1950s in the Soviet Union as a way to train for the Olympics. As the Russians did well at the 1952 Helsinki Olympics, it put periodization on the map as a novel and groundbreaking way to train. Because of this attention, periodization was established as a formal training practice in 1964 by Leo Matveyev. Establishment of periodization as a training practice was primarily the result of Matveyev's book, *Fundamentals of Sports Training* (312). Matveyev's periodization training principles were later adapted for use in Western countries in the mid-1980s. Tudor Bompa is often credited with being the founding father of modern periodization and, more specifically, responsible for bringing the concepts of periodization to Western countries (94, 95).

Specific to running, Arthur Lydiard was a New Zealand running coach who is largely recognized for his periodization model that placed a large emphasis on

base training performed at threshold pace (752). Lydiard also focused on improving running economy through hill running as well as interval and speed training phases (753). As will be discussed later in this module, this approach most closely resembles the *block periodization* model.

Following are discussions of the three primary periodization models: classic, block, and intermediate (hybrid).

CLASSIC PERIODIZATION

The classic periodization model is based on a one-year schedule with typically one or two peaks. The year is broken down into cycles (94).

1. **Microcycle:** Typically represented by a week
2. **Mesocycle:** Consists of two to six micro cycles
3. **Macrocycle:** Refers to the annual training program and is broken into three phases:
 - a. **Preparation:** represented by training done to prepare for the competition phase
 - b. **Competition:** typically designed around one or two competitions and also includes the taper periods
 - c. **Transition (off-season):** used to recover physically and mentally

Figure 13.4 illustrates the classic periodization model.

- **Bottom Line:** microcycles (week)
- **Middle Line:** mesocycle (four microcycles)
- **Top Line:** phases of macrocycle

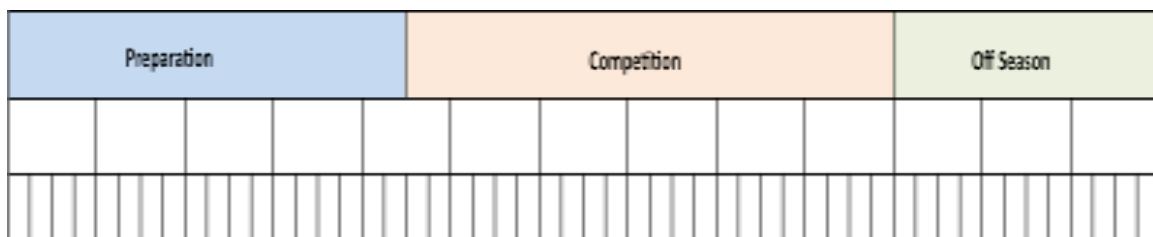


Figure 13.4 Classic periodization model

Classic periodization is often referred to as *linear periodization*. The overall trend of classic periodization is as follows:

1. **General Conditioning:** varied low-level fitness routines
2. **Base Fitness:** sport-specific low intensity, increasing volume
3. **Build Fitness:** integration of higher intensity, volume increases
4. **Peak Fitness:** intensity continues to increase, overall volume may stay the same or decrease – max-distance long run often done in this phase
5. **Taper:** volume substantially decreases, periods of short intensity
6. **GOAL EVENT**
7. **Recover:** typically characterized by the “off-season”; similar to general conditioning

This chart (figure 13.5) represents how these training cycles of linear periodization fit into the classic periodization model.

Preparation	Competitive	Transition
General Conditioning	Peak Fitness	Recovery
Base Fitness	Taper	
Build Fitness	Goal Event	

Figure 13.5 Integration of training cycles into classic periodization model

One of the hallmarks of classic periodization is that several fitness workloads are developed at the same time. Figure 13.6 visually denotes the simultaneous development of multiple fitness workloads. This chart does not accurately represent a training season as it is used solely to represent simultaneous training workloads.

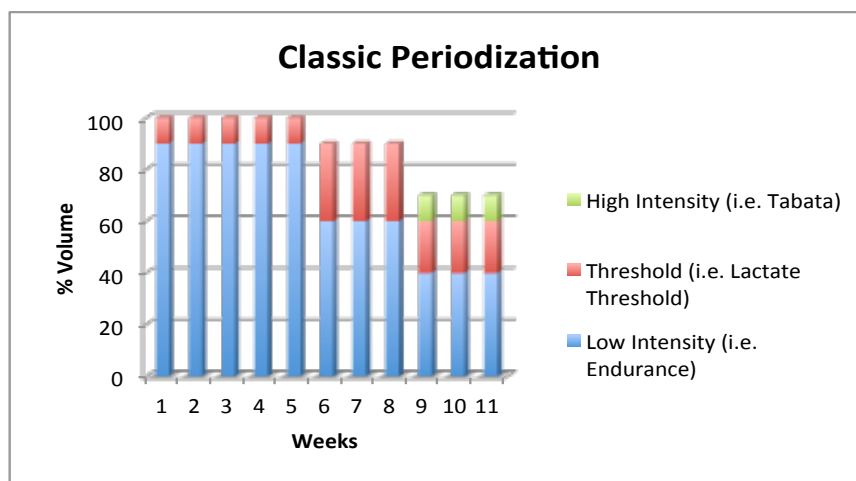


Figure 13.6 Classic periodization: simultaneous training workloads

Criticisms of Classic Periodization

Initial criticism of classic periodization came from Dr. Yuri Verkhoshansky, a sports scientist who is deemed the “father of plyometrics” (233). Dr. Verkhoshansky’s criticism of Matveyev’s classic periodization model resulted in the creation of his own periodization model, *block periodization*, which is discussed in detail in the next section.

Dr. Vladimir Issurin is another detractor of the classic periodization model. Dr. Issurin is a sports scientist, currently working with the Israeli Olympic Committee’s Elite Sports Department (231, 232).

While Dr. Issurin and Dr. Verkhoshansky had many points of contention with the classic periodization model, the points most applicable to endurance sports training are (233):

1. Originally developed for the 1952 Helsinki Olympic Games and constructed around one peak versus multiple peaks
2. Inflexible
3. Too many areas of simultaneous focus that result in insufficient training stimuli and less-than-desired outcomes
4. Incompatible workloads (i.e., high, low) result in conflicting and diminished physiological responses

Of these points of contention, #3 and #4 represent the primary areas of focus. When these two points are summarized, what is being proposed is that classic periodization is equivalent to multitasking. More specifically, too many areas of differing intensity levels are being trained at the same time, effectively negating the desired physiological responses.

While Dr. Issurin is largely critical of classic periodization, he notes in his book *Block Periodization: Breakthrough in Sport Training* that classic periodization is appropriate for beginner to intermediate athletes, but not elite athletes (314).

BLOCK PERIODIZATION

As already noted, the theory of *block periodization* was developed by Dr. Verkhoshansky to correct what he viewed as flaws in the classic periodization

model. The genesis of this model is that multiple areas of focus are not addressed simultaneously, but rather individually over a set period. Block periodization is characterized by a series of connected four-week-long groups or “blocks” (essentially *mesocycles*). Each block has a primary focus.

According to Plisk and Stone, 2003 (230), there are two types of blocks: *accumulation* and *restitution*. During an accumulation block, one area of fitness is emphasized to the extent of intentional overreaching. The point of this is to stress the body with one aspect of fitness (i.e., high volume) over a set time duration – usually four weeks. Toward the end of the training block, fatigue typically sets in because of the extreme workload.

Following the accumulation block is the restitution block. This block looks to benefit from all of the hard work performed during the accumulation block. During this block, the fitness aspect focused on during the accumulation block is greatly reduced while another fitness aspect is introduced at a moderate level (i.e., speed/intensity) (230).

Block periodization is not without its detractors. A 2012 article by John Cissik notes that *USA Track and Field* has not implemented block training because it “tends to lead to the detraining of non-targeted motor abilities” (234).

HYBRID (INTERMEDIATE) PERIODIZATION

Based on the information regarding classic and block periodization, there are two constants:

1. Classic periodization is structured for beginner and intermediate athletes
2. Block periodization is structured for advanced/elite athletes

As these two periodization models (classic and block) are at opposite ends of the spectrum, the question arises: Is there a model that incorporates aspects of both models? Yes, a hybrid (intermediate) model.

Plisk and Stone reference an intermediate periodization model that uses a *summated approach* (230). This approach focuses on fitness versus fatigue and views each as having opposing effects on the body. Therefore, fitness = good, fatigue = bad. A summated approach looks at the sum of fitness and fatigue with

the goal of maximizing fitness and minimizing fatigue. A successful program based on this approach relies heavily on fatigue management practices. A summation periodization strategy is believed to offer dual benefits of reducing the chance for overtraining and increasing fitness as the athlete trains in multiple aspects of fitness in parallel throughout the mesocycle (235, 236, 237).

The intermediate model theorized by Plisk et al. uses four-week mesocycles. During the first three weeks (microcycle) of each mesocycle, the volume or intensity increases each week. The fourth and final week of the mesocycle is an “unloading” or recovery week that is characterized by a decrease in volume and intensity. This allows the body a chance to recover for the following week, which is the start of the next mesocycle. Following are two 16-week sample charts that illustrate the basic concept of the Plisk et al. intermediate periodization model. Note each cycle represents four sequential weeks.

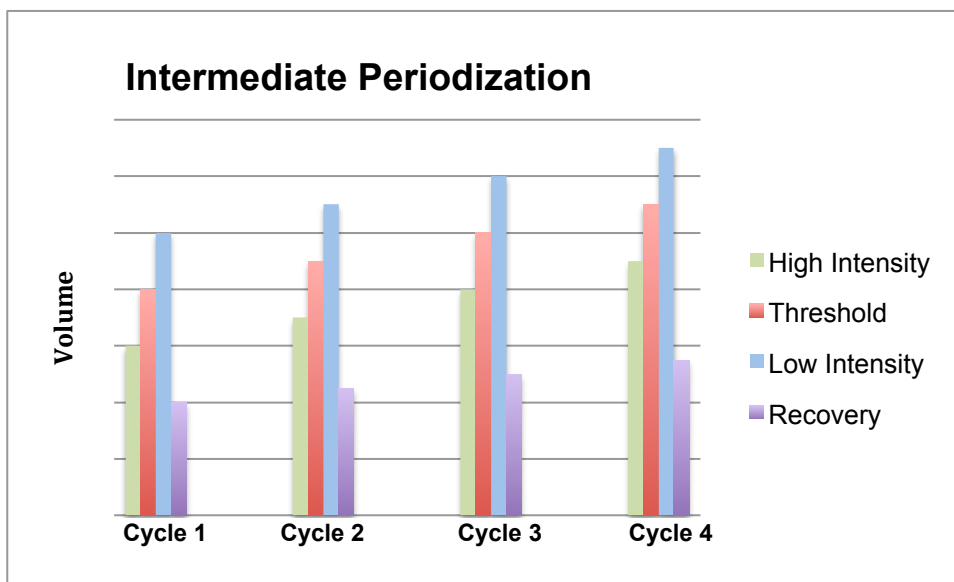


CHART 1

Figure 13.7 Intermediate periodization model: Chart 1

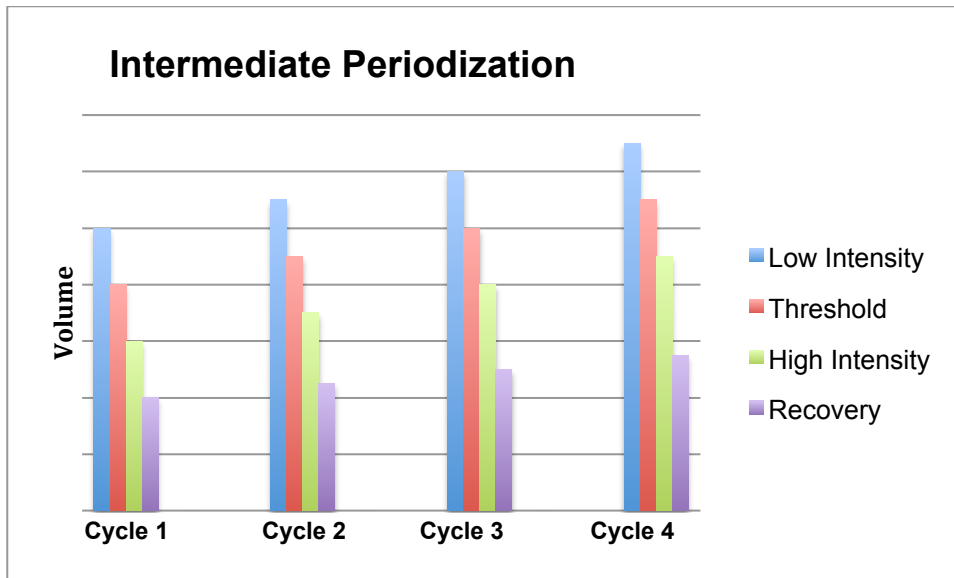


CHART 2

Figure 13.8 Intermediate periodization model: Chart 2

CRITICISM OF PERIODIZATION

The concept of periodization has been widely criticized for being antiquated, rigid, and lacking a scientific and practical foundation. In other words, detractors of periodization believe that it fails to meet the reality of training and competition.

The most applicable criticism of periodization in regard to coaching runners is that periodization is typically viewed and programmed on an annual basis. Therefore clients who hire a coach specifically to train for a singular event do not fall into the annual training window.

Additional criticisms of periodization include (782):

- Not based in science/physiology
- Does not take into account the improvement of athletes and therefore slows the rate of improvement
- Rigid structure that is broken up into predefined periods and cycles
- Complex and confusing principles and guidelines
- Not practical in real-world training applications
- Arbitrary organization

It may be true that if just one type of training was focused on for a set period (i.e., endurance) an individual would see the greatest gains in that area in comparison with training multiple areas of fitness simultaneously. However, fitness elements and, more to the point, training aspects do not exist in isolation. UESCA does not subscribe to the belief that multiple areas of fitness cannot be trained within the same period because of diminished training adaptations. Training in this manner is not efficient or realistic.

UESCA believes that multiple areas of focus should be trained simultaneously to ensure that a client is well rounded and not overtrained in any one area. The most important aspect of a properly constructed training program is that an athlete has individually based stressors placed upon them followed by some duration of recovery. This process is repeated over and over to achieve the desired result.

SUMMARY

Conceptually, periodization is on the right track – to progress an athlete from a relatively low level of conditioning/intensity/volume to a high level in a systematic, progressive manner. This is more or less the genesis and purpose of all training programs. So what's the issue?

The problem is not so much the goal of periodization but rather the means by which an athlete is progressed and the time frame in which the progression occurs.

Perhaps the biggest issue is that periodization forces athletes into the same restrictive training parameters. What if athlete A attains the best results when performing high-intensity training whereas athlete B realizes peak performances when training at or right below lactate threshold? In respect to periodization, if these two athletes were in a mesocycle that focused on below-lactate-threshold training, athlete A would likely realize slower progress in relation to athlete B.

The correct way to progress a client is by taking an individual approach. While it's fine to have a general theme for a training duration (i.e., general fitness), it should not be overly restrictive, meaning, it's fine to allocate the month of March to general fitness. However, the program should not state that every run must be below XYZ heart rate. It's fine to have an overriding theme for a particular

duration. However, it's not realistic, practical, or advantageous to have only one type of training intensity for an extended period. Training guidelines should largely be directed by assessment checkpoints, a client's training history, and the individual's physiological response to training.

Therefore, the proper way to progress an athlete through a training plan is to schedule a baseline assessment and periodic assessment checks (i.e., training volume, physiological assessments, timed distances, etc.) throughout the plan to ensure that an athlete is on target. This ensures that if a client is behind schedule with preparedness, it will be quickly identified and rectified.

❖ PROGRAM ADHERENCE

A training program has value only if it is followed. A primary cause of overtraining occurs when a client adds more volume and/or intensity than a program dictates. Common reasons for a client to add more volume and/or intensity include:

- Client feels good during training so goes past what was scheduled.
- Trains with friend and follows the training program that day.
- Uncomfortable with going as slow and easy as scheduled on rest days.
- Has free time on a scheduled low-volume training day, so the client adds volume.
- Takes advantage of nice weather to do more volume than was scheduled.
- Client thinks that adding greater volume/intensity will lead to faster improvement.

If you think your client has a propensity for overtraining in any of the preceding scenarios, it is advised to address this issue early on. Adherence to a well-designed program has a high probability for success, whereas deviating from a program often equates to failure from the standpoint of proper preparation and success on race day.

Conversely, habitually skipping training sessions or reducing the volume/intensity of workouts will substantially reduce the effectiveness of the program as well.

If clients modify their program, they likely will not tell you. Assuming this is the case, it is up to you to try to determine if this is happening, and why.

Missed Training Days

Your clients will inevitably have unforeseen breaks in their training schedule due to injury/sickness, travel, or work-/family-related issues. Regardless of the reason, many clients (and coaches) freak out at the thought of seeing their hard work and elevated fitness level evaporate into thin air after a few days off.

Periods of inactivity are often referred to as *loss of fitness* or *detraining*. Modifying a training program and to what degree correlates with the number of workouts missed or, more specifically, the total amount of time away from training. The reason why a client missed training days must be factored into the modifications of a training program. If your client missed workouts because of illness, he or she will need to come back at a lower intensity and duration than if training was missed due to nonillness-related issues (e.g., travel, work).

The most common reason for missed workouts is illness/injury.

The most important thing to remember regarding missed workouts is that training cannot be forced. Cramming for an event will increase the chance for overtraining as well as injury. When dealing with missed training days, remember to retain all rest and easy days. Clients respond differently to missing workouts. If your client is new to running, it is likely you will need to spend time reeducating the person on skill development.

If the number of missed days is substantial enough to affect your client's preparation for a goal event, the goal event should be reconsidered. Depending at what point in the training program the lapse occurs, a loss of two consecutive weeks of training would likely put a client at risk of not being in the proper condition to compete in the goal event, or at the very least, the person should have much lower expectations because of a reduced fitness level.

Several studies on this topic examine the effects of training cessation (detraining) on one's fitness level, both in trained and untrained individuals (339, 339, 340). Here are the findings:

1. The largest decrease in aerobic fitness occurs in the first few weeks of the cessation of exercise. The rate of aerobic decline substantially slows after several weeks of detraining.

2. Even after long periods (e.g., 12 weeks) of detraining, VO_2 max was decreased on average only 18 percent in one study.
3. Factors outside VO_2 max, such as muscular efficiency, glycogen levels, and blood enzyme levels, may be influenced greater by time off than VO_2 max.

Based on these studies, it is important to reassure your clients that while their aerobic fitness will decrease, their fitness level will not return to baseline (read: they will not have lost all the hard work they put into training).

The following guidelines for missed workouts correspond to consecutive missed workouts and represent general guidelines that should be interpreted on an individual basis (318). Reduction of workout distances and intensities relate to the originally scheduled volumes and intensities.

Breaks of longer than two weeks from running will likely require an individual to readapt to the musculoskeletal demands of running. Therefore the first few runs back should be short and easy. While individuals typically adapt quickly after restarting a running program, it is common to experience delayed onset muscle soreness (DOMS).

1–4 Missed Days

Consider this a long rest or recovery period. Resume training with no changes.

5–8 Missed Days

Start with two or three easy runs then return to the scheduled training program at the point where a client would have been if the person didn't miss any training. In other words, the client is skipping over approximately one week of training.

9–14 Missed Days

Allow for three or four days of easy running, then resume training at the point where the client ended training. The volume/intensity may need to be slightly reduced.

14+ Missed Days

Allow for four to six days of easy running then resume training at the point where a client ended training. This loss of training will likely affect the overall

training program as well as your client's readiness for his or her event. If you do not believe your client will be ready for the event and that participating in the event could pose a risk, you need to adjust the goal event. If your client is a beginner, it is likely that the person will need to refocus on skill development and form drills.

- These guidelines are in regard to consecutive missed days of training
- If a client missed more than three weeks of training, a period of longer than six days of easy running is likely necessary before increasing intensity/volume

These are rough guidelines. Your client's physiological adaptations and overall energy level should be the guiding principles after missing training days, especially when recovering from illness or injury. In other words, your clients will experience the effects of detraining differently and therefore any training modifications must be solely based on the individual, not on predetermined program modifications.

In regard to marathons, Pfitzinger et al. recommend that if 20 or fewer training days are missed and there are more than eight weeks to go prior to a marathon, the goal race does not have to be revised. However, if there are less than eight weeks before the goal marathon and an individual has missed between 10 and 20 consecutive training days, the goal race should be revised (720).

Vacation/Business Travel

A common issue that interferes with training is travel. Your client may or may not be able to train while traveling. This is especially the case with business travel. Reasons for clients being unable to train while traveling include a lack of time, jet lag, and being in an unsafe area.

Rarely are conditions ideal for training while traveling, and concessions often are necessary. For example, if a gym is not available, bringing along a theraband to do strength training in the hotel room is a common workaround. Another common scenario is not being familiar with the geographic area, so performing cardiovascular training in a gym environment is advised.

Successful integration of training while traveling revolves around planning in advance of the trip. Some preparations you can suggest to your client that may help accommodate training include:

1. Contact hotel ahead of time to determine if a gym is available and, if so, what features it has (e.g., pool, weights, treadmill, etc.)
 - a. inquire about gym hours if your client plans on training early or late in the day
 - b. if hotel does not have a gym, call local gyms regarding a day pass
 - c. if no gym, bring small workout items to use in the hotel room
2. Inquire with the hotel or local running/bike shops regarding safe routes.
3. If traveling with family, have the client design an active vacation such as hiking, swimming, etc.
4. If your client is traveling to an area of high altitude, advise the individual to reduce the intensity/duration of training for the first few days until properly acclimated.

❖ TRAINING PROGRESSION

Overall Themes

When constructing a training program, it is fine to have themes for certain periods. However, the themes don't represent a rigid structure. Themes should not overly dictate the specific workouts, as this is largely determined by the individual needs of a client. Themes provide place markers for clients so that they know where they are in the training program and what the overall focus of a training time period is.

So how does this differ from periodization?

Periodization is a structured program that is broken down into rigid time tier-based phases (macro, meso, and microcycles) of typically 12 months. Conversely, themes serve more as a general overview for different areas within a training program but are not overly restrictive in regard to the content.

NOT A REQUIREMENT

There is no requirement to use themes in a training program. However, they are often a good way to provide guidance and serve as a way to educate clients on their progression and general areas of focus.

Demonstrate Proficiency

Progressing through a training program is similar to moving up through school grades. In order to move up a grade level, there must be a core level of competency in the current grade level. If students do not demonstrate that they are performing correctly to move on to the next grade level, they are typically either held back or given extra help in their weak areas.

Conversely, if an academically gifted client is progressing at a faster rate than his or her classmates, this will also produce a situation that is less than ideal for the student, in terms of inadequate challenge.

This analogy holds true for progressing through the training program. As a coach, you need to assess the level and rate of progression of your client.

A slow progression rate might require reassessment of the goal event.

While there are not levels in a nonperiodization-based training program, it is important to have assessment checks throughout a training process to ensure that a client is on track in relation to the goal event. This is discussed in greater detail in the next section.

Assessing Proficiency

Assessment checks monitor clients to ensure they are progressing and on track for their selected goal event(s). The nature of the checks should be based on the individual. Common assessments include:

- Volume/endurance (e.g., able to run 10 miles)
- Intensity (e.g., able to run at tempo pace for 20 minutes)
- Heart rate (e.g., 25 beats or more heart rate drop in one minute)

- Other
 - Skill-based
 - No discomfort

If a client does not “pass” one or more of the assessment checks, you must determine what modifications, if any, need to be made to the training program. Assessment check goals must be attainable and realistic. Assessments must align with and be plotted in respect to the event(s) being trained for.

The exact locations in a training program where formal assessments occur are individually based. It is advised to not let more than one month pass without some form of assessment. An assessment could be as basic as resting heart rate or as involved as an MLSS assessment.

TRAINING PROGRAM THEMES

Each of the following themes identifies a particular focal point of training (i.e., aerobic development). While there is a focus for each theme, it is normal for themes to overlap so that multiple areas are being focused on at the same time. The below image denotes this overlap.

February	March	April	May	June	July	August	September	October	November
General Fitness									
	Aerobic Development								
			Lactate Threshold						
						Peak			
								Taper	RACE

Figure 13.9 Training program themes

The themes and their placement noted in figure 13.9 are solely for demonstration purposes. If using themes, you can name them as you see fit in ways that best describe the main training focus at a particular time.

The themes noted in figure 13.9 are described in more detail next.

GENERAL FITNESS

This is not running specific but rather focuses on the overall health/fitness of a client. This is placed at the beginning of a program and is practical for clients who are using running to get in shape or lose weight or who are deconditioned.

AEROBIC DEVELOPMENT

Focusing on increasing one's aerobic fitness is akin to building a foundation of a house. You need a foundation before doing anything else such as putting up walls, windows, electrical, etc. Without aerobic fitness, other aspects of training are less impactful and therefore greatly marginalized. An individual who does not have a solid aerobic base typically has a higher heart rate and lower lactate threshold as compared with someone with a solid aerobic base. Without a solid aerobic base fitness level, the quality of training sessions will likely change dramatically from session to session. For example, an individual might feel pretty good in training one day and might feel horrible the next day. To some degree, this is a function of training and will happen regardless. However, by establishing an aerobic base, large day-to-day fluctuations will be greatly diminished.

How exactly does one focus on aerobic training? It's pretty simple: go easy and gradually increase training volume. The primary factor that influences aerobic training is it should be done below lactate threshold. More specifically, an athlete should try to stay within 75 to 90 percent of estimated LT. The following are physiological adaptations that occur during prolonged low-heart-rate training (97, 98, 106).

- Increase in mitochondrial density
- Increase in capillary density
- Increase in heart stroke volume
- Decrease in heart stroke rate

Some degree of aerobic fitness typically carries over from one year to the next. Therefore experienced endurance athletes typically come into a new year of training with a higher level of fitness than those new to exercise.

For some individuals, working out at this low level is very difficult. This is primarily because the effort is not very intense and thus often misinterpreted as being too easy and therefore a waste of time.

While there may be a period in a training program where aerobic development is emphasized, aerobic development-type workouts typically constitute a large part of a runner's training program.

Characteristics

- Low heart rate (below LT)
- Gradual increase in training volume
- Limited intensity

Things to Look For

Your client's resting and recovery heart rate should be declining and his or her intensity level should be increasing without a significant increase in heart rate. A sign that the correct physiological adaptations are taking place would be if a client's speed/pace increases while the intensity level (e.g., heart rate, RPE) stays the same.

LACTATE THRESHOLD

This theme is focused on increasing one's overall intensity/volume. The goal is to increase a client's ability to handle increased stress loads, thereby increasing the individual's fitness level.

Workout types such as long intervals, repeats, and tempo efforts are commonly integrated into this theme. The intensity level of these workouts starts low and increases as the body adapts from both cardiovascular and neuromuscular standpoints.

Characteristics

- Overall increase in intensity from aerobic training. Hard workouts are at or slightly above LT.

Progression Assessment

An individual's LT/VT should increase and the person's heart rate recovery time should continue to decrease. If your client is training for a half or full-distance marathon, the person's overall training volume and, more specifically, the long run days will increase gradually during this time. Overall strength and muscular endurance levels should also increase.

PEAK

This theme represents the point at which your client is doing the longest runs and integrating the highest-intensity workouts. At this point, you should have sufficient data about your client's recovery rate, LT threshold, and adaptation level to the training program. The exact makeup of workouts is based on this data.

Characteristics

- Highest intensity/volume
 - Intensity-based workout sessions typically above LT
- Fine-tune fitness level before goal
- Includes high-volume days

TAPER

The definition of taper is: a gradual decrease (315). This more or less sums up what a running taper is all about. To arrive at an event as prepared and fresh as possible, some form of a taper is recommended. The longer the event, the more accurate this is.

Without a taper, the chance that an athlete would be exhausted and perform well below potential is highly likely. The purpose of a taper is to allow all of the hard training that has been performed throughout a program to be realized on the day of a race. The goal of a taper is to decrease fatigue without losing fitness. Rest is just as, if not more, important than physical exertion. Without rest, the body breaks down and performance decreases. A taper is not followed by a training phase but rather by the event for which the runner is training.

So what type of increase in performance can be expected by tapering? Based on research studies, the range of improvement is between .5 to 6 percent, with 3 percent being the average performance increase (685).

Characteristics

- Decrease in training volume
- Reduction of the duration of intensity, while retaining intensity level

Reasons for Taper

There are five primary reasons why tapering is an essential component of a successful training program (100):

1. Increase glycogen stores
2. Repair micro tears in muscles and connective tissue
 - a. Multiple studies found that tapering reduced muscle damage and increased muscle rebuilding/remodeling (680, 684).
 - b. Exercise-associated muscle cramps are often associated with long-distance running. A 2011 study by Schwellnus et al. found exercise-associated muscle cramping was influenced by prerace muscle damage (681).
3. Increase mental freshness
4. Decrease chance for injury and sickness
 - a. A 2013 study by Papacosta et al. found that a taper period between one and three weeks resulted in reduced physiological stress and increased immune function (682).
5. Increase red blood cell and total blood volume

Structure and Duration of Taper

While most runners agree that tapering is an important part of race preparation, there is not a universally agreed upon way to do it.

Taper Guidelines

While the durations for each individual will vary, these are some guidelines for a running taper:

- Gradually reduce training volume.
 - More rest days (do not decrease training frequency by more than 20 percent) (685).
 - Decrease in volume typically ranges from 10 to 60 percent in total (685).
- Maintain intensity but reduce duration and frequency of hard workouts.
 - Reduce the overall work performed while maintaining fitness and the neuromuscular response from intense training.
 - Increase time between intervals (i.e., increase rest periods).
- The duration of a taper is in direct proportion to the distance of the event being trained for. Below are the standard taper durations for the four primary race distances. The specifics of a taper must be customized to each individual. The durations below are averages and therefore can be shortened or lengthened accordingly.
- Get plenty of sleep.
- Stay hydrated and fuel properly.
- Those with a high level of fitness and older clients will typically require a longer taper (318).
- Tapering is all-inclusive. Tapering is not just about a decrease in running volume but rather a reduction in total energy expenditure.

Marathon: Two to four weeks
Half Marathon: One to two weeks
5K/10K: Two days to one week

Figure 13.10 Taper durations

While the taper duration range of marathons is between two and four weeks, two and three weeks are the most common. This is likely because this time duration allows an athlete to reap the maximum benefits from the taper while not becoming detrained.

While 5K and 10K events typically do not have a structured taper, in a study by Houmard et al. it was found that a seven-day taper for a 5K event resulted in improved running performance and running economy (679).

- Aside from taking off or running easy the day before, many well-trained, experienced runners often do not taper for short-distance events such as 5K and 10K races. This is because their ability to recover from hard efforts is typically better than less experienced runners and those with lower fitness levels.

Taper Structure

There are two primary taper types (388):

1. **Step:** Characterized by a singular reduction in training volume
2. **Exponential Decay:** Characterized by a steady decrease in training volume (nonlinear reduction in volume)

Quite a few studies have looked at the benefits and drawbacks of step versus exponential decay tapers (389, 390, 391, 392, 393, 394, 685). The general findings:

- Exponential decay tapers elicit increased performance over step tapers.
- Step tapers typically occur over a longer period than exponential tapers.
- One day of complete rest (no exercise) per week should be integrated into tapers.
- No decrease in the level of intensity during the taper. However, the duration of intense workouts and the quantity of intensity sessions decrease throughout the taper.
- A long taper might decrease performance on race day.
- The most intense overload training should be done the week before the taper.
- Do not decrease mileage during taper more than 90 percent.

- The positive effects of a taper last only so long. After this point the physiological effects of detraining begin. Studies have shown this point exists between two and four weeks but is based on the individual.

Taper Intensity

As stated above, intensity is still part of the program during a taper, but in limited quantities and durations.

Taper Variables

There are three primary reasons why taper distances and times vary among individuals:

1. **Client Recovery Rate:** The ability of a client to recover will influence how long or short a taper should be.
2. **Volume:** The greater the volume of a training program, the greater the length of the taper. This typically correlates with the distance of the event being trained for.
3. **Current Fatigue Level:** The more fatigued a client is, the longer the taper. However, if your client is substantially fatigued prior to the taper, it is likely the person is overtrained.

While the reasoning behind a taper is based in science, determining the exact taper schedule relies on a lot more than purely science. As with one's performance on race day, there is a fair amount of luck involved with determining the perfect taper. However, by adhering to the proper training progression and taper, individuals put themselves in the best position possible to be successful on race day.

Despite tapering being a generally accepted training principle, many runners find it hard to commit to following a taper plan through its entirety. This is commonly referred to as *breaking the taper*. Two common reasons for this are:

1. **Afraid of losing fitness:** If a taper is done properly, this is not a cause for concern. As long as there are bouts of intense exercise integrated into a taper, there will be no loss of fitness. Additionally, muscular and

cardiovascular benefits of exercise are not realized until at least seven to 10 days after an exercise session.

- 2. Type A personality:** The same reasons why your client is superintense and very driven might also be why it's hard to adhere to a taper program. While your client might have become involved with running for the hard and long workouts, this is at odds with what occurs during a taper. It is your responsibility as the coach to remind clients why they need to taper and keep their focus on the goal. Clients need to respect the integrity of the training process.

One important thing to be aware of and to communicate to clients is that they may feel bad during the taper. This usually occurs a third or halfway through the taper (101). This typically applies only when training for a marathon. This is because the training volume for marathons is substantially higher than that for other running events. The reduction in training volume during a taper has a profound effect on the body. The body basically “shuts off” as it responds to the decrease in training volume. This is normal and expected. It is crucial to make your client aware of this because not doing so could result in your client freaking out and breaking the taper. You need to reassure clients that this is a normal physiological response and that they will feel better as the taper progresses.

Creating a Taper Schedule

To determine a taper schedule, you must first determine how long the taper will be. This is often a tricky proposition for the following reasons:

To determine the exact length of the taper phase, you should first determine your client's ability to recover. To a large degree, this can be ascertained only by witnessing physiological responses to the training program. This is at odds with determining the length of the taper because, in order to structure a training program in totality, the length of the taper typically needs to be taken into consideration at the beginning of the program – before you know exactly what your client's recovery ability is. Therefore, in order to structure the taper properly, you must assume from the beginning that your client will need the maximum taper duration and then reduce the taper if needed.

A meta-analysis of multiple taper studies across multiple sports (running, swimming, cycling, triathlon, rowing) found that a two-week, exponential taper

with 41 percent and 60 percent volume reductions (without altering training intensity/frequency) elicited the best results (686). Specific to running, this guideline most closely identifies with the half marathon. While this guideline should not be adhered to for all running distances, it does provide a general framework from which one can adapt it to varying race distances.

Figure 13.11 shows suggested taper reduction percentages based on the week prior to the taper, which is typically the highest-volume week in a training program.

Taper Reduction Schedule			
	5K/10K	Half Marathon	Marathon
Week 1	30–50%	40%	20–30%
Week 2		60%	40–60%
Week 3			60–75%

Figure 13.11 Taper reduction schedule (percentages)

- The above taper schedule is a rough guideline, and as with all other areas of programming, the taper needs to be customized to the individual needs of your client.

Race Distance-Specific Taper Tips

Taper Intensity Workouts

The most common intensity workouts are repeats, fartleks, and short-tempo runs/efforts. The idea is to perform bouts of high intensity in relatively small quantities (ex: 5K at race pace, five 800-meter repeats at race pace) but not on consecutive days. These workouts are often integrated with low-intensity running (ex: run one mile easy, one mile at race pace, one mile easy).

5K/10K

- Reduce or eliminate a long run the week prior to the race (723).
- Take the day off or do a recovery run the day before the race.

Half Marathon

- During the first taper week, limit runs to six miles.
- During the second taper week, target running two or three times with the max mileage run being four miles.
- The last two days should be very low key. Performing a few strides, doing a recovery (shakeout) run, or taking the day(s) off is an option. The exact “exercise prescription,” or lack thereof, should be based on the way the client best responds to recovery structure (i.e., day off, shakeout run, etc.). This is determined earlier in the training program.
- During the first week of the taper, perform a dress-rehearsal run (three to four miles: two miles at race pace). Mimic race day in regard to clothes, shoes, and fueling, and ideally run at the same time of day as the race. This will help prep the mind for the upcoming race!

Marathon

- While overall volume is reduced week-by-week, high-intensity days are still a part of the program.
 - According to Pfitzinger et al., this is as much for psychological reasons as physical. This reminds runners that they are still fit and therefore gives them confidence (520).
- During the first week of the taper, limit the long run to 15 miles.
- During the second week of the taper, limit the long run to 10 miles.
- No runs over five miles during the final taper week.
- The last two days should be very low key. Performing a few strides, doing a shakeout run, or taking the day(s) off is an option. The exact “exercise prescription,” or lack thereof, should be based on the way a client best responds to recovery structure (i.e., day off, shakeout run, etc.). This is determined earlier in the training program.

POST-RACE

After a race, and specifically a longer race such as a half or full marathon, it is normal to have a period of rest. Unlike a taper, the rest period not only has minimal training volume but also minimal intensity. The purpose of a rest period is to allow the body and mind to recover sufficiently to continue the training process.

The longer an event is, the longer the rest period typically will be. A 5K or 10K might have just a day or two of rest post-race while a half or full marathon might have five to seven days rest post-race, respectively.

Many clients who hire running coaches are not looking for an annual program. These clients typically are looking to train for one or two running events. Maybe they continue with running, maybe they don't. This is mentioned because oftentimes your tenure as "coach" comes to an end after an event. This does not mean that you cannot and should not advise them on the proper amount of rest post-race and how to maintain their newfound fitness level.

OFF-SEASON

For runners who train and compete year after year, the off-season represents a period where the overall volume and intensity is much lower than the racing season. For some runners, this is a welcome respite from the rigors of training, whereas others despise it and are champing at the bit to compete and get back to a full training load.

Off-season training greatly influences the performance of the upcoming racing season. The off-season is a time to take stock in how the past season went and, based on this information, focus on strengthening weak areas, identify a racing calendar, and allow the body to rest and recover – both mentally and physically.

There is no set duration for the off-season. However, it is suggested that it be no shorter than two months as the body needs this amount of rest to recover fully from the racing season (assuming a full season was undertaken).

Many runners gain weight during the off-season or when they are not actively training for an event – this is normal. Your client should continue to eat a healthy diet and exercise a minimum of three or four days per week during the off-season. Runners who gain too much fat weight during the off-season often find themselves fighting to lose this weight throughout the racing season the following year. While the off-season is a time to relax and eat/drink some of the things that are limited during the season, moderation is advised.

One of the biggest benefits of the off-season is the opportunity for a runner to mentally recharge and find new motivation. Training for running events requires a lot of focus, commitment, and energy. Without an off-season or periods of reduced training intensity/loads, many runners would find themselves burned out mentally.

As your clients are most likely not professional/elite athletes, you need to understand that their motivation for running is likely predicated on improving their health and fitness as much as on the opportunity to compete. Therefore, when discussing off-season training with your clients, you need to inform them that keeping a focus on their overall fitness level is important.

Characteristics

- Substantial reduction in training volume and intensity
- Mental break
- Rest
- Skill/form work
- Goal setting/structure training program for following season
- Cross-training
- Assessing strengths/weaknesses
- Trying out new equipment/clothing/shoes

The duration and level of activity during the off-season is greatly influenced by the proposed competition schedule the following year.

PEAKING

Peaking is a systematic process by which an individual reaches the highest possible level of physical conditioning on a set date or time frame.

Structure is an important aspect to peaking successfully. Peaking too soon or too late will negatively affect the performance of a client on race day.

The duration an athlete can stay in peak condition varies greatly between individuals. Generally speaking, the more aerobically fit an athlete is, the longer

the individual can remain in peak condition. Even if an athlete could stay at peak physiological condition for a relatively long period, the person might purposefully look to reduce peak condition in order to prepare and peak for another event later in the season.

Tapering is an integral part of the peaking process.

HISTORICAL DATA TO PREDICT PEAK

Peaking is an area in which experienced runners have a distinct advantage over novice runners. The more times athletes peak for events, the more they learn each time. The ability of an athlete and/or coach to learn from each peaking experience is contingent upon the ability to dissect and analyze the training process and associated race performances. Assuming an athlete was assessed via quantitative measures throughout the training process in prior seasons, the individual can compare this data against current assessment data. Cross-checking this information is a great way to assess progress or lack thereof.

As noted previously, formal assessment checks throughout the training process are important to ensure that a client is on the right track. Informal assessments should be done daily and weekly by getting client feedback on the training and looking for trends such as poor recovery or mental burnout. This data should be used not only for the race currently being trained for, but also for future races and race seasons, if applicable.

Experience Counts

Peaking at the right time is very much like hitting the bull's-eye on a target, as there is a very small margin for error. Like hitting the bull's-eye, through repeated attempts at peaking, one can adjust strategy to increase the likelihood of peaking at the correct time.

REALIZE PEAK POTENTIAL

While some athletes can stay in great condition for a long time, to truly peak for an event requires changes in both training intensity and volume. The best way to

understand this would be to think of an athlete who is in good physical condition for the whole season and consistently operates at around 85 percent of potential. This is in contrast to an athlete who peaks for an event. Ideally, this athlete would hit at or near 100 percent of potential for an event, but at other times during the season the person might operate at only 75 percent of potential.

This illustrates how, in many ways, peaking for an event is a gamble. As a coach, you are gambling that your peaking strategy will garner your client a great result by being at or near 100 percent of potential on race day. However, if the peaking strategy is incorrect and the person can realize only 80 percent of potential on race day, the client potentially could have had a better performance by not peaking and just trying to maintain a relatively consistent high level of fitness throughout the season. So, the question is: Are you aiming for 85 percent or 100 percent from your client? Hopefully, it is the latter.

The following graph illustrates the aforementioned athletes, one who stays at 85 percent of potential (athlete B) versus the athlete who peaks for events (athlete A). The purpose of this is to illustrate that athlete A stands a better chance of performing at top level than athlete B.

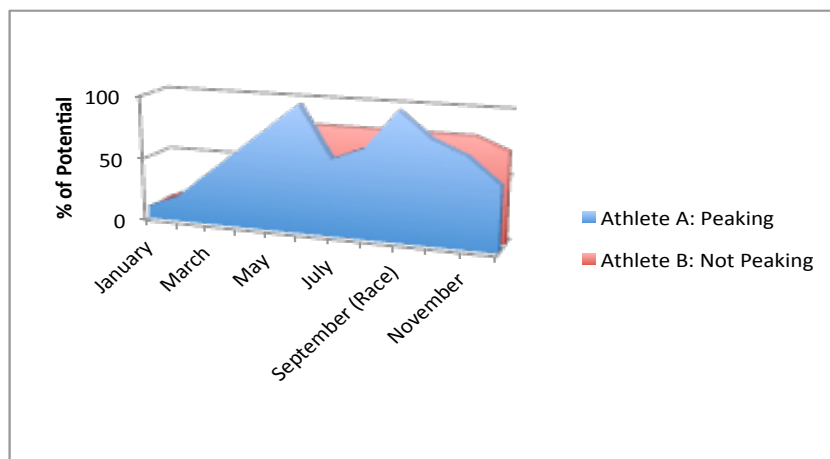


Figure 13.12 Athlete comparison: peaking

It is not possible or plausible to stay at peak form for long periods. The peak should represent a relatively short period followed by a decrease in intensity. Remember, recovery is when the body is given the opportunity to recuperate from hard efforts and, therefore, make progress.

As noted earlier, many athletes think that they can stay at top form (i.e., peak) for longer than they actually can. It is natural to think this way. Your client has put a

ton of work and time into getting to peak and, because of this, the individual wants to try to stretch out this peak for as long as possible. Trying to extend the time spent at one's physiological peak will likely result in injury and/or diminished performance down the road. The change between peak and off-peak fitness happens quickly. For example, an athlete can put in a stunning performance one week and the next week performance can be greatly diminished.

MULTIPLE PEAK EVENTS

Peaking for an event is a gradual process that takes a minimum of several months. Because of this, it is likely not possible to properly and successfully peak for more than three distance events per year (i.e., marathon). The spacing of events is critical, and there should be no less than two months between events for which a client peaks. The less aerobically fit an athlete is, the more time is required between peak events. When looking at a typical racing season, a two- and three-event peak schedule might look something like this.

Three-Event Peak: March, June, October

Two-Event Peak: June, September

Following are several reasons why someone may not peak at the right time:

1. Injury/Illness
2. Lack of training
3. Attempt to peak too many times during the season
4. Break taper

Even if clients have peaked properly for an event, they could have an “off day,” meaning they are unable to perform to the full extent of their current fitness level.

Troughs

The time between peak events should be viewed as micro seasons. After a peak event, there is a rest period, followed by an increase in distance and intensity, followed by a taper, prior to the next peak event. The period between peak events (i.e., *troughs*) are made up of a combination of training intensity/volume.

The exact training durations and intensities during a trough are primarily based on the needs of the client and the duration between peak events.

Training through Events

If your client is using events prior to the goal event for preparatory purposes, the individual should not be in peak condition for these events.

Unless you inform your client of this, it is easy for the client to worry over not being in shape or behind schedule with training.

OVERTRAINING

Overtraining occurs when one's performance decreases and/or the person becomes injured because of volume/intensity that is too high. Overtraining often correlates with a lack of sufficient rest and recovery. In other words, overtraining occurs when one's training stress exceeds ability to recover.

Training for distance running takes a lot of time – especially for marathons and ultra-marathons. This often leads many individuals either to train too little or too much. Finding the sweet spot between the two increases the likelihood of an optimal performance when race day arrives. While undertraining does happen, this section will focus on clients who train too much and lack the proper amount of rest and recovery.

While there are varying ways to avoid the negative effects of overtraining, none is more effective than a properly structured and executed training plan with the primary goal of not ending up in the exhaustion phase of GAS.

It is difficult to identify what overtraining looks like as everyone has a bad day from time to time – and even a string of bad days is not uncommon. Symptoms of overtraining include a halt of forward progress and overall fatigue for long periods (longer than is normal with a particular client). When you have identified what you think is overtraining, the most important thing to do is modify the program to allow the client to recover both physically and mentally. If you ignore signs of overtraining and continue to push a client, the individual will continue to see performance decline both physically and mentally.

While some studies have purportedly identified a workload in regard to the “correct” number of intense, endurance, and recovery days per week, the reality is that there is no universally correct ratio of hard/easy days that minimizes the chance of overtraining, as this is individually based.

The body adapts to stress. As a result, a workload today that may elicit symptoms of overtraining may be the correct workload a month or two down the road.

Symptoms and Prevention

Common symptoms of overtraining are:

- Lack of energy
- Loss of motivation and enthusiasm for training and racing (i.e., mental burnout)
- Compulsive need to train and exercise
- Substantially elevated/depressed heart rate during exercise and at rest
- Inability to recover from workouts
- Decrease in training capacity and intensity
- Increase in injuries
- Consistent muscle soreness

Some of these symptoms can be indicative of serious medical issues. Therefore, if after adjusting your client’s program, the person still exhibits overtraining symptoms, a medical checkup is advised.

So what should you do if you suspect your client is overtraining? Aside from revising the program to allow for more rest both physically and mentally, it is also helpful to emphasize these other areas:

- Sports massage
- Hydration
- Cross-training

Following are several ways to prevent overtraining:

1. Track heart rate during both exercise and resting
2. Keep a daily training log that includes mood, diet, and rate of perceived exertion (RPE)

3. Implement a properly constructed training program
4. Cross-train
5. Incorporate enough recovery into the training program
6. Increase training distance, intensity, and time progressively and intelligently
7. Ensure your client is properly hydrated and maintains a healthy diet
8. Stress the importance of sleep in the training process and have clients track the number of hours slept each night as well as the quality of sleep

Capacity for intensity, duration, and mental stress varies substantially from individual to individual. What might result in overtraining for one client may not result in overtraining for another.

Multiple Factors

While a program itself might not lead to overtraining, many other factors may result in your client experiencing the same symptoms as overtraining. Some of these factors may be stress or a lack of sleep. A program does not have to be superintense for the symptoms of overtraining to be present. It can simply be a function of your client dealing with many life issues that ultimately take a toll on the body and mind, resulting in the individual being unable to adequately recover and progress.

Unless You're a Pro, Don't Train Like One

Despite the many books and magazine articles on how to train like a professional runner, most of us do not have the physical capacity to train like a professional, and more important, most of us do not have the time. You will not find too many people who would argue that training four to five hours a day with the right structure would elicit great results. However, when is the last time you heard of a nonprofessional athlete who had this amount of time to allocate to training daily?

Have Fun!

There are 168 hours in a week. If 56 hours are allocated for sleeping (eight hours/day) and 56 hours for work (eight hours/day), this leaves 56 hours (eight hours/day) for everything else, including training, family time, breakfast, dinner, work events, etc. Get the point?

With so much on your client's plate, it is easy to forget to have fun from time to time. This can lead to mental burnout. During recovery periods, the focus should be on resting and having fun. Whether this means going out with friends after work or going to the movies, your client needs to continue to enjoy life and not allow running to become the singular focus.

REST

Because of the many training aspects related to running, especially marathon training, it is easy to understand how one can overtrain and minimize or eliminate rest days all together. As noted previously, if an individual trains for too long or too intensely without rest, the person will end up in the exhaustion phase. In this phase, all training benefits are negated until the individual is able to rest and recover.

The physical act of training is only half of the equation to becoming a better runner; the other half is rest and recovery.

Trial and Error

The amount and type of rest will vary by the individual. Some clients might respond best to a total day off while others might respond best to active rest (e.g., an easy bike ride or swim). Additionally, some clients may respond best to two consecutive days off followed by two hard days, whereas others may respond best to just one day off followed by one hard day.

Unless your client already knows this and passes this information along to you, the only way to find out is through trial and error. Tracking all this information, including sleep hours, will help you understand how your client most efficiently recovers.

While this certification discusses rest in regard to rest days or active rest days, rest should be viewed as a function of energy expenditure and muscle activation. Just because your client is not training on a rest day does not necessarily mean the person is resting. For example, spending the day at a concert standing in one place for hours or taking a four-hour walking tour of NYC are not rest days! If your client is serious about training, the person needs to be thinking about energy conservation at all times.

The more stress placed on the body – whether from racing, training, lack of sleep, or long work hours – the more rest is required. Additionally, each individual's recovery time will vary based on many factors, such as injury, fitness level, muscle soreness, etc.

Hours, Not Days

It is advantageous to think of training days in terms of hours, not days. Let us say your client gets off work at 9 p.m. and therefore trains from 10 p.m. to midnight. The next day your client has an after-work dinner, so the person trains in the morning from 5:30–7. While the training sessions are technically on consecutive days as the training program states, only 5.5 hours passed between the end of one training session and the start of the next. This is not a sufficient amount of time between training sessions and, in this case, certainly not enough time for adequate sleep. It is ill-advised to perform evening-to-morning workouts on consecutive days or same-day workouts that are of high intensity and of a similar nature (strength to strength or cardiovascular to cardiovascular).

❖ SUMMARY

- *Periodization* is based on the principles of the *general adaptation syndrome* (GAS).
 - The three phases of GAS are:
 - Alarm
 - Resistance
 - Exhaustion
- *Classic periodization* consists of three phases: preparation, competitive, and transition.
- *Classic periodization* is broken down into cycles:
 - Macrocycle
 - Mesocycle
 - Microcycle
- *Classic periodization* is criticized by some for being too general and trying to focus on too many training variables simultaneously, thereby reducing a program's effectiveness.
- *Block periodization* is applicable only to elite athletes.
- A *hybrid periodization* structure combines elements of both classic and block periodization.
- While a training program must have structure, it also must be flexible.
- Your client will miss training days. You may need to modify the program to accommodate this.
- Training themes can be implemented to provide guidance and nonrigid structure to a training program.
- UESCA does not believe that periodization in its formal state is applicable to most athletes as it is too restrictive and therefore not practical to use in real-world training applications.
- Tapers are characterized by gradually reducing volume while still integrating some intensity.
- It is ideal to include assessment checks throughout a program to ensure a client is progressing properly.
- Peaking for an event gives your client the best chance to perform at physiological maximum on the day of the event.
 - For experienced athletes, historical data can help to estimate the peak with greater accuracy.
- Finding the right balance between exercise, rest, and other obligations (e.g., job, family, etc.) is critical for a successful program. Individuals who

do not successfully balance the demands of training can become mentally burned out and suffer symptoms of overtraining.

- Think of training days in terms of hours, not exact days.
- Overtraining occurs when the training volume and/or intensity exceeds the ability to recover.
- The off-season offers an opportunity for a well-deserved mental break.